Voyage economics and the time charter equivalent

What a dry bulk or tanker voyage earns a day, the freight it needs, the costs that decide it, and the physical fixture against the FFA, on the Baltic Exchange's routes in 2026

Research note by Nathan Couturier, 29 October 2026. Interactive site: https://nathancouturier.github.io/voyage-tce-model/. Code and data: https://github.com/nathancouturier/voyage-tce-model.

Summary

- A VLCC fixed on TD22, Galveston to Ningbo, at the Baltic's USD 58.22 million in the week to 23 October 2026, earns about USD 466,378 a day over a 110.5 day round voyage; freight of USD 17.24 a tonne would only cover its voyage costs. The Baltic's own round voyage TCE that week was USD 451,905; the engine reads 3.2 percent higher, about the port charges it leaves out.
- On routes outside Europe, bunkers are every sourced dollar of voyage cost at the August 2026 market; on voyages into or out of the EEA, EU allowances at the 2026 phase in are 12.6 to 17.4 percent of it (EUA EUR 82.31 a tonne). A Hormuz war risk premium of 7.5 percent of a USD 157 million VLCC hull would cost USD 11.78 million a transit, 6.5 times TD3C's bunker bill.
- Round the Cape of Good Hope an LR2 on TC20 sails 31.5 more days than by Suez and needs USD 10.92 million (USD 121.33 a tonne) more freight to earn what Suez earns with the Red Sea premium paid; Suez stays the better way until its toll passes USD 6.58 million.
- For a VLCC from the Gulf, a 7.5 percent Hormuz premium needs USD 45.31 a tonne more freight to keep its TCE; with 20 days of waiting to transit, at the TCE of the week to 23 October 2026, USD 145.17.
- In the week to 3 July 2026 a C3 cargo at USD 30.11 a tonne earned USD 38,791 a day, USD 141 a day below the July 2026 Capesize 5TC FFA; a C5 cargo at USD 12.25 earned USD 44,723 a day, USD 5,791 above it. The FFA stood USD 5,254 a day above the spot 5TC: the market priced the coming months above the week's spot.
- Higher freight raises the speed that maximises the TCE and dearer bunkers lower it, on every one of the routes tested (C5, C3, TD22, TC20).

A TCE here is an estimate of what a voyage earns at stated inputs, not a forecast, and it is set against operating and capital costs, never net of them.

Of USD 58.22 million of freight, USD 51.56 million is left after commission and USD 4.48 million of voyage costs, all of them bunkers. Over 110.5 days (54.9 in ballast, 50.7 laden and 5.0 in port), the result is a TCE of USD 466,378 a day.

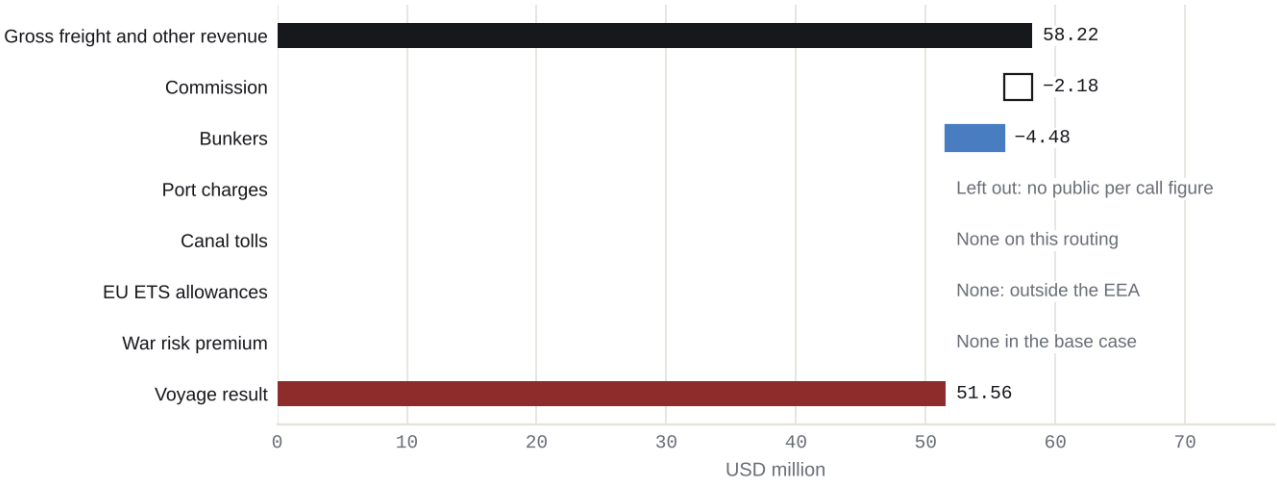


Figure 1. Engine output for TD22 in the week to 23 October 2026. Freight: Baltic Exchange via Hellenic Shipping News. Bunkers: Star Asia Shipbroking weekly market report, week 43 of 2026, 23 October 2026. A cost with no public figure is left out and named in its row. Retrieved 29 October 2026.

1. The question and the method

For a given ship, route and market: what does the voyage earn a day, what freight does it need, which costs decide the answer, and is fixing the cargo physically better than hedging the same exposure on paper? The study covers the routes the Baltic Exchange defines in its Guide to Market Benchmarks v8.8 (September 2026): Capesize C3 and C5, Panamax P7 and the Supramax trip S4A for dry bulk; TD3C, TD22, TD20 and TD25 for dirty tankers; TC2, TC14, TC1 and TC5

for clean, with TC20 and TD34 where the Baltic's lump sums and the Hormuz bypass need them.

The TCE is gross freight and other revenue, less commission and voyage costs, over the days of the whole round voyage, the ballast leg included: TCE = (freight + other revenue − commission − voyage costs) / voyage days. Voyage costs are bunkers by sulphur zone (MGO inside emission control areas, measured on the route line), port charges, canal tolls, EU allowances and war risk premiums. The tanker TCE follows the Baltic's own method (Appendix 3 of the guide): a 5 percent weather margin on steaming days, the round voyage from the discharge port, the port and waiting days and consumptions of the route's description, and bunkers at the hub nearest the load port. Worldscale freight is WS / 100 x flat rate; flat rates are licensed and none is public for 2025 or 2026, so a Worldscale rate is priced only with the reader's own. Commission is 5 percent on dry routes and 3.75 percent on tankers.

Every input is either sourced (a dated public document, with the passage it was read from), derived (a stated rule on sourced values) or empty: a figure with no public source stays out of the TCE and the result names it. Port charges have no public per call figure for any study port and Suez and Panama tolls depend on each ship's canal tonnage, so every TCE here leaves them out and reads high by their amount. A dated market reads no price from after its date. The engine is a Python package, mirrored in the browser and held to it within one part in a billion on 275 cases.

2. Validation

The Baltic publishes its tanker routes on a Worldscale and a TCE basis. Only two study routes are assessed as a lump sum in dollars, TD22 and TC20, so only they can be priced without a licensed flat rate: 5 weeks between 27 February 2026 and 23 October 2026, with bunker prices of the same week, reproduce the Baltic's TCE within 3.2 to 5.2 percent, the engine always above it by about the port charges it leaves out. The 64 Worldscale anchors cannot be reproduced without a licensed flat rate, and are recorded as such rather than tuned. The Baltic's own worked TD3C calculation of July 2017 is reproduced line by line: laden 19.85 days, ballast 20.64 days, 46.49 days in all, and 1,389.20 tonnes of fuel laden; it prints no prices. Searoute distances fall within 2 percent of NGA Pub. 151 for 17 of 21 port pairs; the largest miss is Ras Tanura to Cape of Good Hope (+3.6 percent).

Early March 2026: Platts assessed the Gulf to China VLCC rate at USD 62.07 a tonne on 2 March 2026, which the engine turns into USD 319,741 a day with the bunkers of the week of 27 February 2026 (USD 255,921 with the early March war risk premium of 2.5 percent of a USD 120 million hull). The USD 89.00 a tonne widely reported on 3 March 2026 names no primary assessment; it would give USD 468,621 a day. The Baltic's TD3C TCE was USD 209,550 in the week to 27 February 2026 and USD 485,959 in the week to 6 March 2026.

The engine reads above the Baltic's TCE every week, by 2.4 to 5.3 percent, except the week to 13 March 2026 (23.6 percent), priced with the bunker table of the week before prices jumped after the closure of Hormuz.

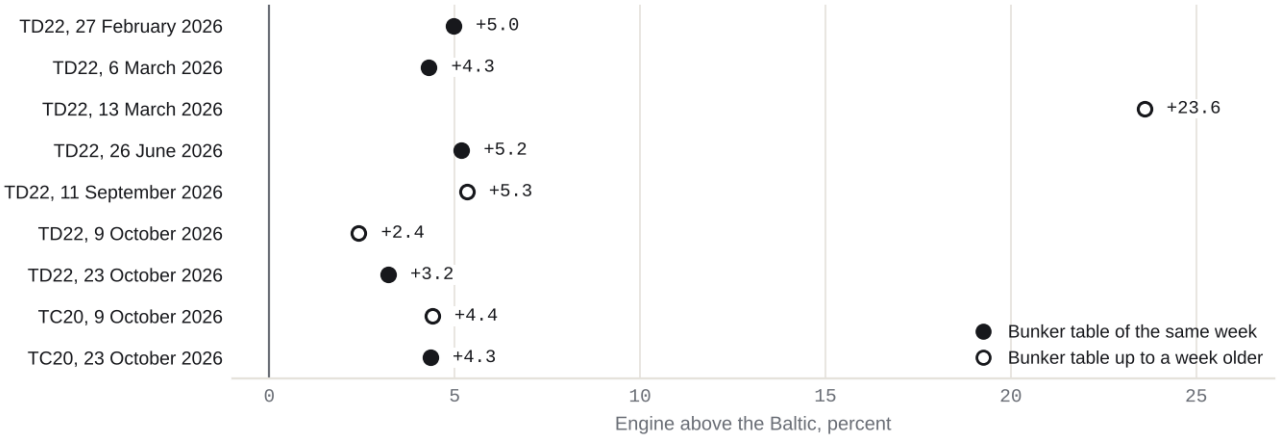


Figure 2. Baltic Exchange weekly tanker reports as republished by Hellenic Shipping News; bunkers: Star Asia Shipbroking and Banchero Costa weekly tables; engine output. Retrieved 29 October 2026.

3. Where the money goes

At one common dated market, August 2026 (Ship & Bunker's monthly averages, the EU allowance at EUR 82.31 a tonne and 1.1594 dollars a euro), sourced voyage costs are bunkers and, on voyages that touch the EEA, EU allowances. The share of MGO in the bunker bill follows the miles inside emission control areas and the days at EU berths.

Table 1. Sourced voyage costs at the August 2026 market, by route.

Route	Voyage	Days	Voyage costs, USD	Bunkers, %	EU ETS, %	MGO in bunkers, %
C3	Tubarao to Qingdao	76.5	2,238,517	100.0	0.0	0.0
C5	West Australia to Qingdao	29.0	870,671	100.0	0.0	0.0
P7	US Gulf to Qingdao grain	87.0	1,483,467	100.0	0.0	4.6
S4A	US Gulf trip to Skaw-Passero	42.6	806,116	84.6	15.4	35.6
TD3C	Middle East Gulf to China	47.0	1,816,997	100.0	0.0	0.0
TD22	US Gulf to China	110.5	3,945,099	100.0	0.0	10.4
TD20	West Africa to UK-Continent	36.3	1,150,605	82.6	17.4	31.0
TD25	US Gulf to UK-Continent	40.0	1,268,237	84.5	15.5	48.1
TC2	Continent to US Atlantic coast	28.3	605,358	87.4	12.6	69.9
TC14	US Gulf to Continent	41.2	827,097	84.8	15.2	47.3
TC1	Middle East Gulf to Japan (LR2)	51.4	1,221,620	100.0	0.0	0.0
TC5	Middle East Gulf to Japan (LR1)	51.9	1,100,833	100.0	0.0	0.0

Engine output; bunkers: Ship & Bunker, August 2026 averages (quarterly where no monthly figure exists); allowances: DEHSt (German auctions, proxy), monthly average for August 2026, at the 2026 phase in; dollars: Federal Reserve H.10, mean of the 21 published days of August 2026. Port charges and canal tolls have no public figure and are not in the costs. Retrieved 29 October 2026.

4. Speed

The TCE was computed at every laden speed on a 0.01 knot grid from 1 knot below the slowest published consumption to 1 knot above the fastest, at the lowest, median and highest public freight of 2026 and at the lowest weekly, the August 2026 and the highest weekly bunker price at the route's hub. The textbook result holds on every route tested: freight pushes the best speed up, bunkers push it down. The guide gives tanker consumption at two speeds only, so the fitted exponent is low and, at 2026 freight, the best tanker speed sits at the top of the range: the curves cannot say whether that speed is reachable.

Table 2. Laden speed that maximises the TCE, knots.

Route	Low freight	Median freight	High freight	Cheap bunkers	Dear bunkers	Fitted exponent
C5	12.0	12.0	13.0	15.0 (edge)	11.0	2.43
C3	12.0	13.0	14.9	15.0 (edge)	12.0	2.43
TD22	13.9	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	2.18
TC20	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	1.60

Freight columns at the August bunker price, bunker columns at the median freight. Edge: the best speed is at an end of the range searched. Engine output on the Baltic's published speed and consumption pairs (Guide to Market Benchmarks v8.8).

5. Detours: Suez, the Cape and Hormuz

Asia to Europe. An LR2 on TC20, Jubail to Rotterdam at the Baltic's lump sum of USD 21.51 million in the week to 23 October 2026, earns USD 334,929 a day by Suez with the Red Sea premium paid, before any toll, and USD 210,416 round the Cape, 31.5 days and 0.43 billion tonne miles more. A Panamax from Qingdao to Rotterdam needs USD 62.92 a tonne to earn the P4 trip rate by Suez, USD 66.89 by Panama and USD 70.29 round the Cape; at the Cape's freight, Suez could bear USD 462,219 of tolls and waiting, Panama USD 213,205.

Hormuz. For a VLCC from Ras Tanura, the premium and the waiting outweigh every routing choice. Loading by a ship to ship transfer off Fujairah shortens the voyage by 2.7 days, and loading at Mina Al Fahal outside the strait (the Baltic's TD34) by 4.8, but the cost of the transfer and of the shuttle through the strait has no public figure. The Baltic's TD34 TCE was USD 656,587 a day in the week to 23 October 2026, against USD 1,287,851 on TD3C.

Per tonne of cargo, the Cape costs an LR2 more than the Red Sea premium does, and the Hormuz premium and waiting outweigh every routing choice for a VLCC.

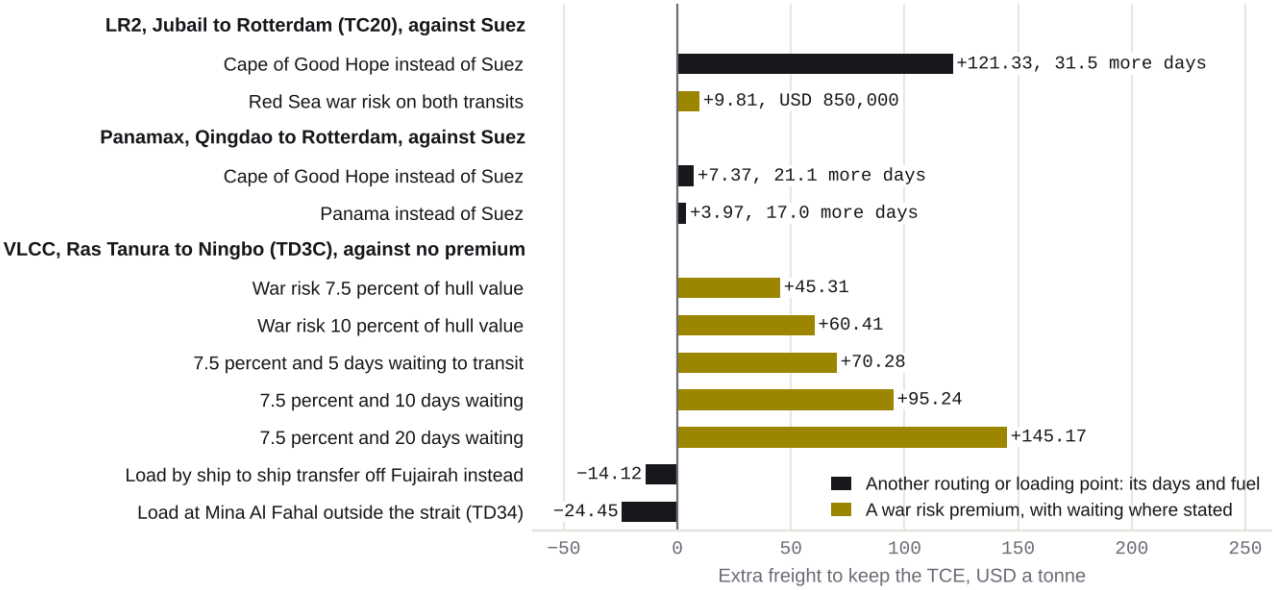


Figure 3. Engine output, week to 23 October 2026: TC20 at the Baltic's lump sum, the Panamax at the freight that earns the P4 trip rate, TD3C at the Baltic's TCE of the week (USD 1,287,851 a day). War risk: Marsh and Platts, July 2026. Suez and Panama tolls have no public figure for these ships and are left out. Retrieved 29 October 2026.

6. Carbon

EU allowances cover half the emissions of a voyage into or out of the EEA and all of one between two EEA ports, for 40 percent of 2024 emissions, 70 percent of 2025 and 100 percent from 2026, when methane and nitrous oxide join carbon dioxide (Directive 2003/87/EC, Articles 3ga and 3gb). Under the BIMCO ETS Allowances Clause for time charter parties the charterer provides the allowances for the ship's emissions during the charter: on the S4A trip the charterer pays USD 66,690 and the owner USD 57,491 for the ballast leg back. FuelEU Maritime's penalty is assessed on a ship's yearly energy, not a voyage's, so it is shown beside the TCE and kept out of it.

Table 3. EU ETS cost of one voyage by phase in year, at the August 2026 allowance price.

Voyage	2024	2025	2026	2026, USD a tonne of cargo
TC23, between two EEA ports	USD 12,348	USD 21,609	USD 31,343	1.04
TD20, into the EEA	USD 78,791	USD 137,885	USD 200,059	1.54
TD3C, outside the EEA	USD 0	USD 0	USD 0	0.00

Engine output at EUR 82.31 a tonne (DEHSt (German auctions, proxy), monthly average for August 2026) and 1.1594 dollars a euro (Federal Reserve H.10, mean of the 21 published days of August 2026); emission factors of Regulation (EU) 2015/757 Annex I as amended, global warming potentials 28 for methane and 265 for nitrous oxide. Retrieved 29 October 2026.

7. Physical against paper

Time charter FFAs such as Capesize 5TC settle on the month's average of the daily index in dollars a day; voyage route FFAs such as C3 and C5, and the tanker routes on ICE, on the month's average in dollars a tonne. A voyage is hedged with 5TC days equal to its days, or with route FFA tonnes equal to its cargo: 76.5 days of 5TC or 170,000 tonnes of C3 for the C3 voyage of the week to 3 July 2026. The basis between a fixture and its index can be small: a C5 cargo fixed on 22 October 2026 at USD 14.15 a tonne for 5 to 7 November 2026, against a C5 index of USD 14.10, earns USD 279 a day more than a cargo at the index. Public FFA quotes in dollars a tonne for tanker routes were not found.

The voyage TCEs sit near or above the 5TC FFA for their month, and the FFA above the spot 5TC: the market priced the coming months above the week's spot.

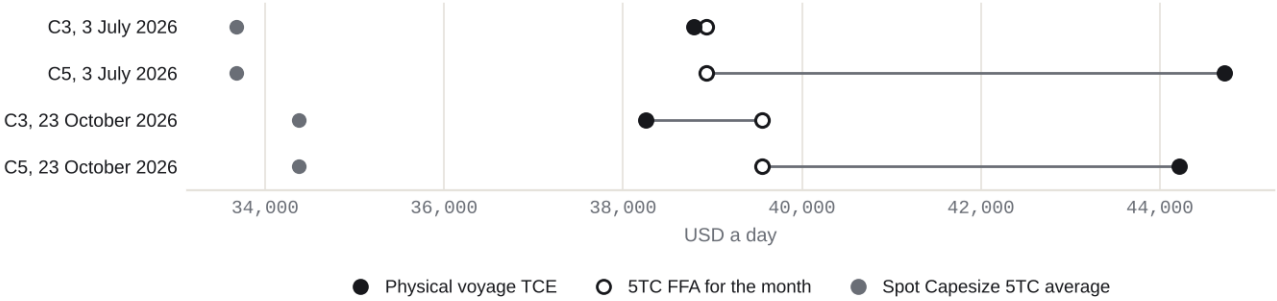


Figure 4. Engine output for C3 and C5 at Banchero Costa's route assessments and the week's bunkers; FFA: Banchero Costa, Dry bulk FFA assessments, 6 July 2026 and 25 October 2026; spot 5TC: Signal Group and Banchero Costa. The July spot figure does not state its vessel basis (180,000 or 182,000 dwt). Retrieved 29 October 2026.

8. What the study cannot do

- Port charges have no public per call figure: every TCE leaves them out and reads high by their amount, USD 1,307 a day on C3 and USD 3,450 a day on C5 for each USD 100,000 of charges.
- Worldscale flat rates are licensed and none is public for 2025 or 2026, so a rate in Worldscale points cannot be priced without the visitor's own flat rate, and those anchors are not reproduced.
- Suez and Panama tolls depend on each ship's own canal tonnage and are left out; the detours report the toll each routing could bear instead.
- Bunker prices are dated public averages or weekly broker tables at four hubs, not the price a ship actually paid; the week of the closure of Hormuz shows how fast they moved.
- FFA levels are dated public assessments, two dry curves; no public tanker FFA quote in dollars a tonne was found.
- The Baltic's tanker speeds and consumptions come as two pairs, so the fitted exponent is low and the best speed often sits at the top of the range searched.
- A TCE is an estimate of what a voyage earns at stated inputs, not a forecast, and it is compared with operating and capital costs, never net of them.

Sources and reproduction

Baltic Exchange, Guide to Market Benchmarks v8.8 (September 2026) and its TD3C TCE calculation process (July 2017); the Baltic's weekly tanker reports as republished by Hellenic Shipping News, December 2025 to October 2026; Ship & Bunker monthly analyses and quarterly tables; Star Asia Shipbroking and Banchero Costa weekly reports; Signal Group and Xclusiv Shipbrokers weekly reports; Marsh and S&P Global Platts as reported in July 2026; EIA, Today in Energy, 26 March 2026; the European Commission's auction reports and DEHSt's German auction reports; the Federal Reserve's H.10; EUR-Lex (Directive 2003/87/EC as amended by Directive (EU) 2023/959, Regulation (EU) 2015/757 as amended, Regulation (EU) 2023/1805); IMO resolutions MEPC.328(76), MEPC.361(79) and MEPC.392(82); NGA Pub. 150 and Pub. 151; searoute 1.6.0 over Eurostat SeaRoute; EEX and ICE Futures Europe contract specifications. Every figure's document, retrieval time and hash are in the repository's data/raw/manifest.json; licensed documents are cited, never redistributed.

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